



S.B. Patil College Of Science and Commerce

Pinprai Chinchwad Education Trust's
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UDISE NO: 27232001412 College Index No J 11 16 065

ACA/DI/21	Prelim 2 Examination	Academic Year :- 2021-22
Rev:00		Total Marks:- 50
Class:- 12 th Science		Date:- 29/1/2022
Time:- 8:30am – 11:30am		
Subject :- Computer Science 1		

Instructions: 1) All questions are compulsory. 2) Figure to the right indicates full marks 3) Draw a neat diagram wherever necessary. 4) Header files and comments in C++ program are not necessary.

Q1)A) Select correct option from the following and rewrite sentence:

- a) ----- tag is used for subscript in HTML code. 1
 i) <SUP> ii) <Subscript> iii) <SUB> iv) <Super>
- b) Finding location of a given element is called as ----- 1
 i) Traversing ii) Inserting iii) Searching iv) None of these
- c) Ability to take more than one form is called as ----- 1
 i) Inheritance ii) Polymorphism iii) Class iv) None of these
- d) ----- is not a visibility label. 1
 i) public ii) private iii) protected iv) reserve

Q1)B) Answer any two of the followings:

- a) Give the structure of HTML web page. 3
 b) Explain any three types of control structures used for flow of control. 3
 c) What is constructor and destructor? State difference between them. 3

Q2)A) Answer any two of the followings:

- a) Explain call by value and call by reference by giving an example for each. 3
 b) What is Binary Tree? Draw the tree structure for the following expression: 3

$$E = (A - B) / ((C * D) + E)$$

 c) Show the representation of records in memory considering suitable example of three records and three fields. 3

Q2)B) Answer any one of the followings:

- a) Write an algorithm for traversing and inserting an element to a Linear Array each of separately. 4
 b) What is operator overloading? State the three steps involved in the process of overloading an operator. 4

Q3)A) Answer any two of the followings:

- a) In case of data structure, explain the following terms: 3
i) Linear Array ii) Linked List iii) Tree
- b) What is class? Give a general form of a class declaration. 3
- c) What is inheritance? Mention the different types of inheritance with suitable diagram. 3

Q3)B) Answer any one of the followings:

- a) What is Binary Tree? Explain terminology describing family relationship between the elements of a Binary tree with suitable example. 4
- b) Write the Binary Search Algorithm with a suitable example. 4

Q4)A) Answer any two of the followings:

- a) What is record? How it differs from a Linear Array? 3
- b) What are different file mode parameters in C++? 3
- c) Write any six features of Object Oriented Programming. 3

Q4)B) Answer any one of the followings:

- a) What are object? Describe how members of a class can be accessed using objects of that class. 4
- a) What is operator function? Describe syntax of an operator function. Explain difference between operator function as a member function and as a friend function. 4

Q5) Solves any two of the followings :

- a) Write a program in C++ to read a set of numbers from keyboard and find out Largest Number in the given array using pointer. 5
- b) Write a program in C++ to display a series of first fifteen terms of Fibonacci Series (0,1,1,2,3,5,-----). 5
- c) Write the HTML code for the following : 5

V.I.P. SALES		
STATIONARY	ERASER	5
	PENCIL	7
	PEN	10
	BOOK	22
TOTAL		44

OR

Solves any two of the followings :

- a) Write a program in C++ to convert the given binary number into decimal equivalent using method convert (). 5
- b) Implement a class rectangle. Each object of this class will represent a rectangle accepting its length and width as float. Include an area function which will calculate the area of the rectangle. 5
- c) Write HTML code to display the following nested list: 5
- College
 - I. Arts
 - a. History
 - b. Politics
 - c. Languages
 - i. English
 - ii. Marathi
 - II. Science
 - a. Physics
 - b. Chemistry
 - c. Biology
 - III. Commerce
 - Accounts
 - SP
 - Economics

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UDISE NO: 27252001412

College Index No: J 11 16 566



ACA/DI/21	Prelim 2 Examination	Academic Year :- 2021-22
Rev:00		Total Marks:- 50
Class:- 12 th Science		Date:- 31/1/2022
Time:- 8:30am - 11:30am	Subject :- Computer Science 2	

Instructions: 1) All questions are compulsory. 2) Figure to the right indicates full marks
3) Draw a neat diagram wherever Necessary.

Q1)A) Select the correct alternative and rewrite the following:

- a) Microcontroller 8051 have ----- external interrupts. 1
 i) 1 ii) 2 iii) 3 iv) 4
- b) The invalid register pair of 8085 microprocessor is ----- 1
 i) BC ii) DE iii) HL iv) SP
- c) PSW is a combination of ----- registers. 1
 i) M & F ii) H & F iii) L & F iv) A & F
- d) A device used for modulation and demodulation process in network is _____. 1
 i) Hub ii) Router iii) Modem iv) Repeater

Q1)B) Answer any two of the followings:

- a) Write a note on evolution of Microprocessor. 3
- b) Explain the any 3 addressing modes of 8085 Microprocessor with one example each. 3
- c) Differentiate between Microcontrollers 8051 and 8052. 3

Q2)A) Answer any two of the followings:

- a) What is multiplexed BUS in 8085? Give its advantages. 3
- b) Explain following instruction of 8085 microprocessor : 3
 i) CMA ii) RRC iii) STC
- c) Define Topology. Explain Physical and Logical Topology. 3

Q2)B) Answer any one of the followings:

- a) What is flag? Explain flags status in 8085 with diagram and example. 4
- b) Explain following terms related to Pentium Processor: 4
 i) Dual Pipeline ii) Branch Prediction iii) On Chip Cache iv) 64bit Data Bus

Q3)A) Answer any two of the followings:

- a) Explain any 3 features of 8085 Microprocessor. 3
- b) What is microcontroller? Write application of microcontroller. 3
- c) Draw a block diagram of ALU organization and explain it in detail. 3

Q3)B) Answer any one of the followings:

- a) Draw memory map diagram of 8051 microcontroller? Explain it in Detail. 4
- b) Explain Contention and Polling Access Methods: 4

Q4)A) Answer any two of the followings:

- a) Explain Coaxial Cable in detail. 3
- b) The register B contains data 69H and register A contains data 49H. The memory location 3900H contains data 56H. After execution of the following instructions independently, What is content in accumulator.
i) MOV A, B ii) MVI A, 97H iii) LDA 3900H 3
- c) Flag register contains C5H. Interpret its meaning. 3

Q4)B) Answer any one of the followings:

- a) What is interrupt? List Hardware Interrupts according to Priority. Explain maskable and Non maskable interrupts. 4
- b) Explain the following characteristics of Transmission media :- 4
i) Installation Difficulties ii)EMI iii)Bandwidth iv)Attenuation

Q5) Solves any two of the followings :

- a) Write an ALP to transfer first 10 bytes of memory block starting from 5000H to a new block starting from 5020H. 5
- b) Write an ALP to count the occurrences of the data 9CH in a memory block starting from memory location 4000H to 400F. Stores the count at memory location 4500H. 5
- c) Write an ALP to subtract the number stored in memory location 3601H from number stored in memory location 3600H. Stores the Positive result at memory location 3602H. 5

OR

Q5) Solves any two of the followings :

- a) Write an ALP to find out 2's complement of five numbers stored from memory location 3330H and onwards. Stores the result from memory location 4100H. 5
- b) Write an ALP to find the product of two numbers stored in memory location C005H and C006H. Stores the result in Memory location C000H and C001H. 5

c) Fill in the blanks for following 8085 instructions :

5

Label	Instructions
START	LXI H , 3900H
	MVI M , 49H
	MOV D , M
	MVI E , 18H
	LDA 3900H
	STA 4500H
	XCHG

- i) D = -----H ii) E = ----- H
- iii) HL = -----H iv) A = ----- H
- v) [4500H] = -----H

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